

Project Design Phase Solution Architecture

Date	20 June 2025
Team ID	LTVIP2025TMID58466
Project Name	ResolveNow – Your Platform for Online Complaints
Maximum Marks	4 Marks

Solution Architecture:

1. Client (Browser / Frontend)

- Built using React.js, HTML/CSS, and Bootstrap/Material UI
- Pages include:
 - Home, Register/Login
 - Complaint Form
 - Live Chat / Video Call (WebRTC)
 - User Dashboard (to track complaint status)
 - Admin Dashboard (to assign and manage complaints)

2. Web Server (Backend API)

- Developed using Node.js with Express.js
- RESTful APIs:
 - POST /register – Register a user
 - POST /login – Authenticate user and issue token
 - POST /complaints – Submit a complaint
 - GET /complaints/:id – Get complaint details
 - PUT /assign – Admin assigns complaint to agent
 - Real-time endpoints via Socket.IO for chat

3. Real-Time Services

- Socket.IO used for real-time messaging and agent-user communication
- WebRTC (optional integration) for video calling

4. Database Layer (MongoDB)

- Stores:
 - User profiles (admin, user, agent)
 - Complaint details (status, timestamps, attachments)
 - Chat history (if persisted)

- Assignment records & admin logs

5. Authentication & Security

- **JWT (JSON Web Tokens)** for login/session security
- **Role-based access control:** user, agent, admin
- **File upload validation and input sanitization** to prevent malicious input

6. Notification System

- **Email or SMS alerts when:**
 - **Complaint is submitted**
 - **Status is updated**
 - **Agent is assigned**
- **Uses services like Nodemailer or third-party APIs (e.g., Twilio for SMS)**

Example - Solution Architecture Diagram:

